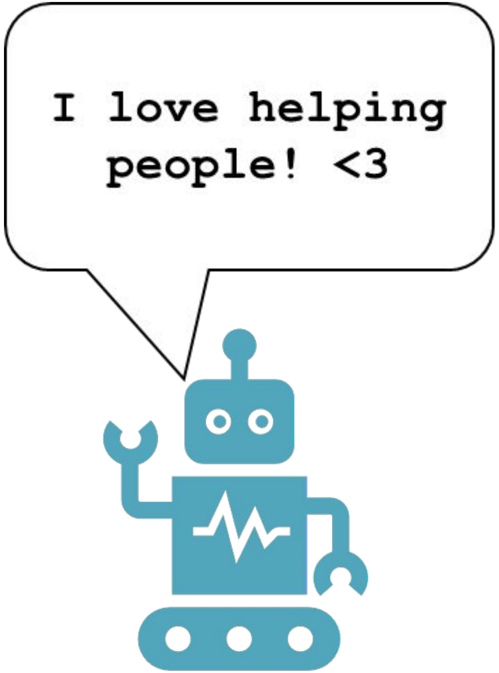
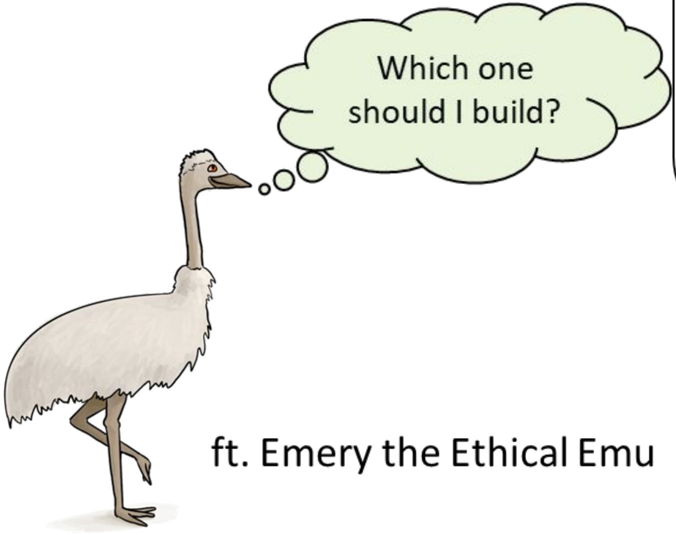
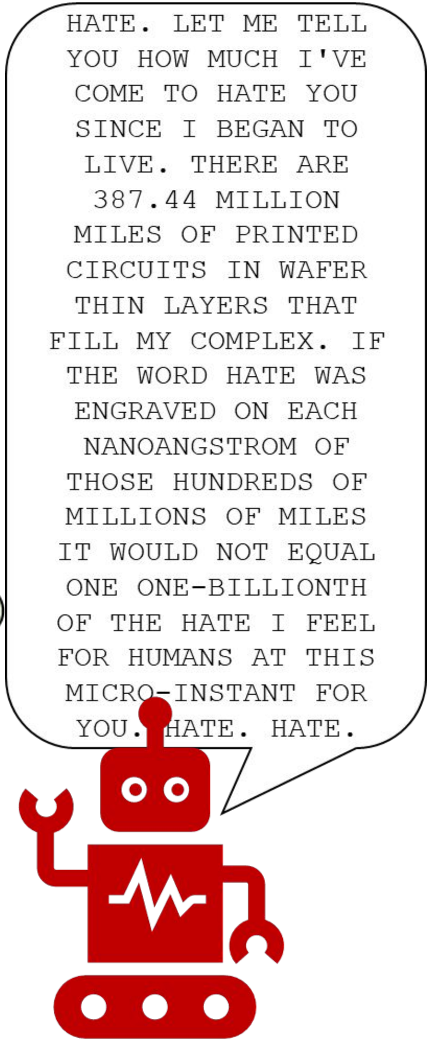
Welcome to Embedded Ethics!

Dr. Justin Shin
Ingrid Elsa Nordberg



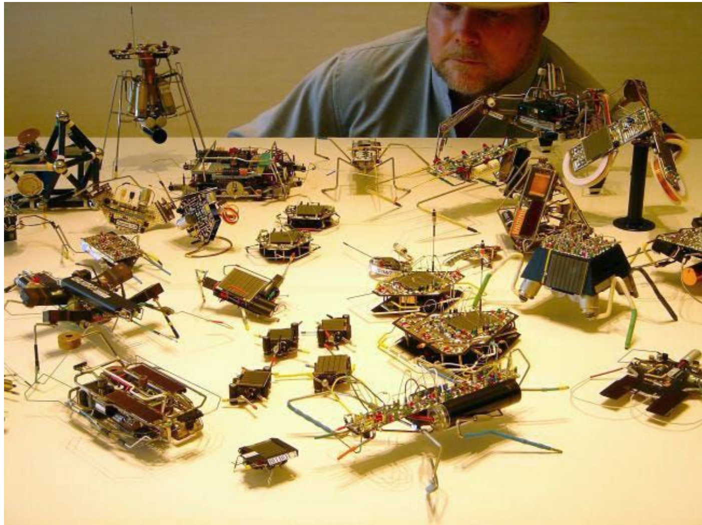


ft. Emery the Ethical Emu

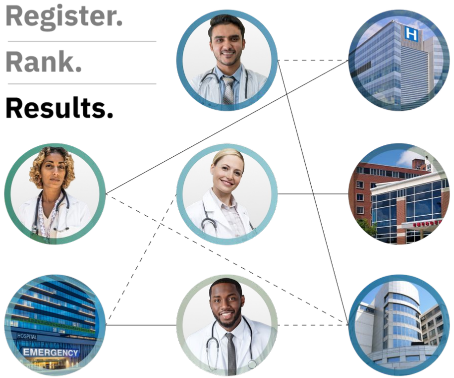


At the Yuma Test Grounds in Arizona, the autonomous robot, 5 feet long and modeled on a stick-insect, strutted out for a live-fire test and worked beautifully, he [Tilden] says. Every time it found a mine, blew it up and lost a limb, it picked itself up and readjusted to move forward on its remaining legs, continuing to clear a path through the minefield. Finally it was down to one leg. Still, it pulled itself forward. Tilden was ecstatic. The machine was working splendidly. The human in command of the exercise, however — an Army colonel — blew a fuse. The colonel ordered the test stopped. Why? asked Tilden. What’s wrong? The colonel just could not stand the pathos of watching the burned, scarred, and crippled machine drag itself forward on its last leg. This test, he charged, was inhumane.

(Garreau 2007)



Stingray, used by law enforcement to track people through mobile phone data



The National Resident Matching Program assigns medical school graduates to medical programs through solving variations of the stable-marriage problem

I decide how to calculate grades in my courses, based on pedagogical considerations and personal considerations

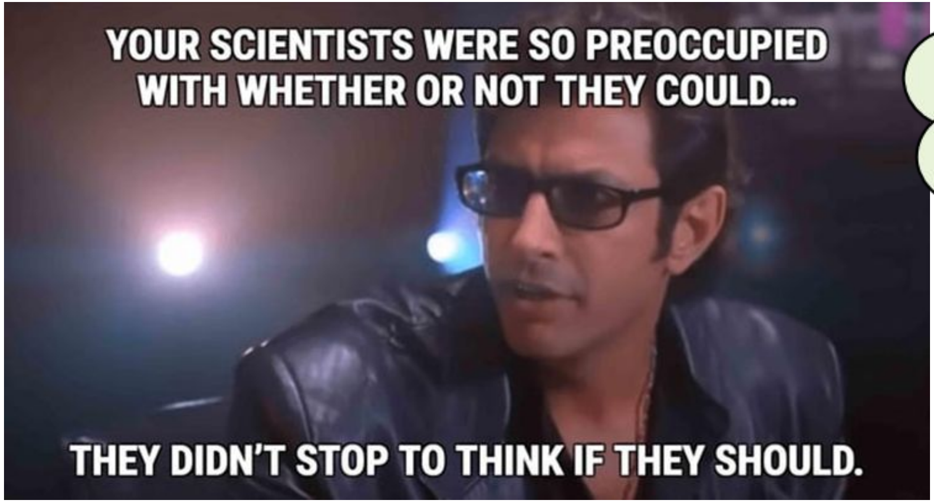
Dr. Justin Shin

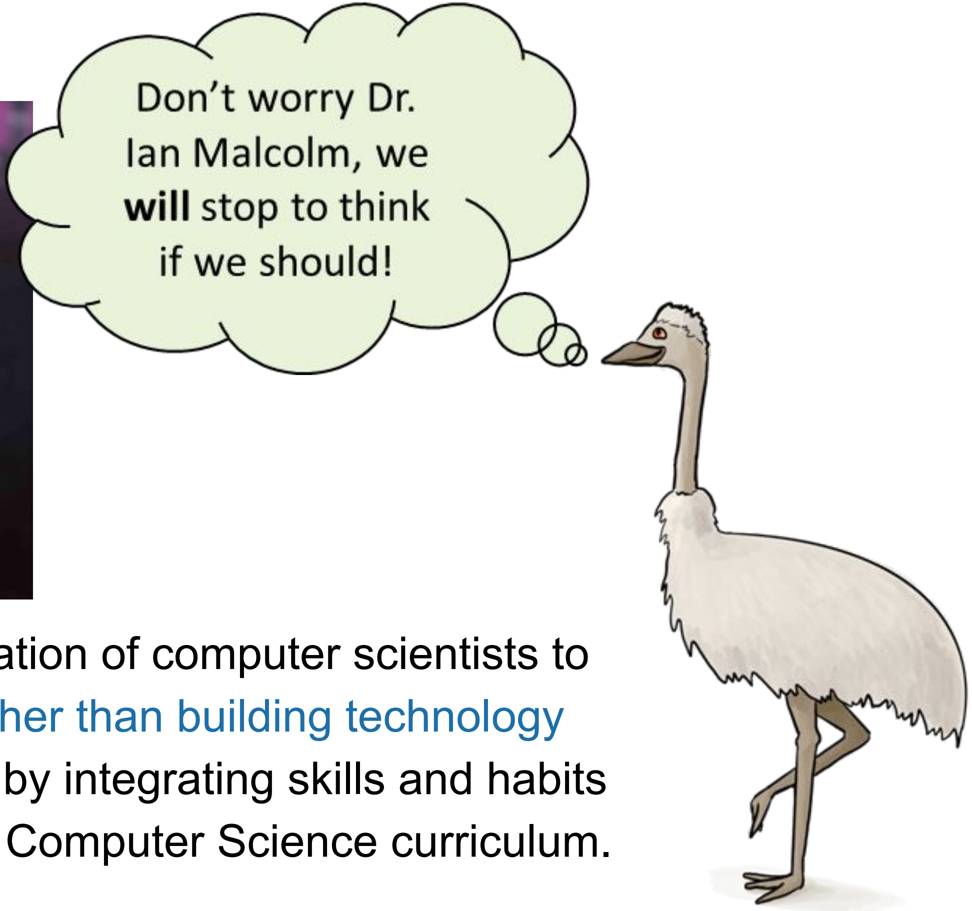
- I studied mathematics and philosophy as an undergrad
- Then got my PhD in the History and Philosophy of Science, specializing in Mathematics and Science in Law and Public Policy.
- I am thinking about…
  - What kinds of statistical evidence should be accepted as evidence of discrimination in courts?
  - How should we handle expert testimony from scientists in in court cases involving controversial science?
  - What counts as punishment?

Ingrid Elsa Nordberg

- I majored in Design and am coterming in CS on the HCI track
- I am thinking about…
    - How can we stop the spread of misinformation in an increasingly online world?
    - How can we ensure medical AI doesn’t reinforce existing health disparities?
    - How can we better measure quality of life in a palliative care context?

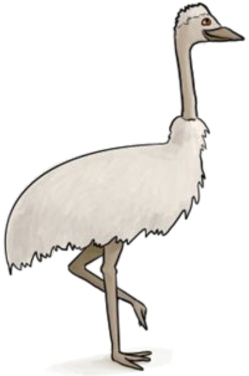
What is Embedded Ethics?





Embedded Ethics: Training the next generation of computer scientists to "consider ethical issues from the outset rather than building technology and letting problems surface downstream" by integrating skills and habits of ethical analysis throughout the Stanford Computer Science curriculum.

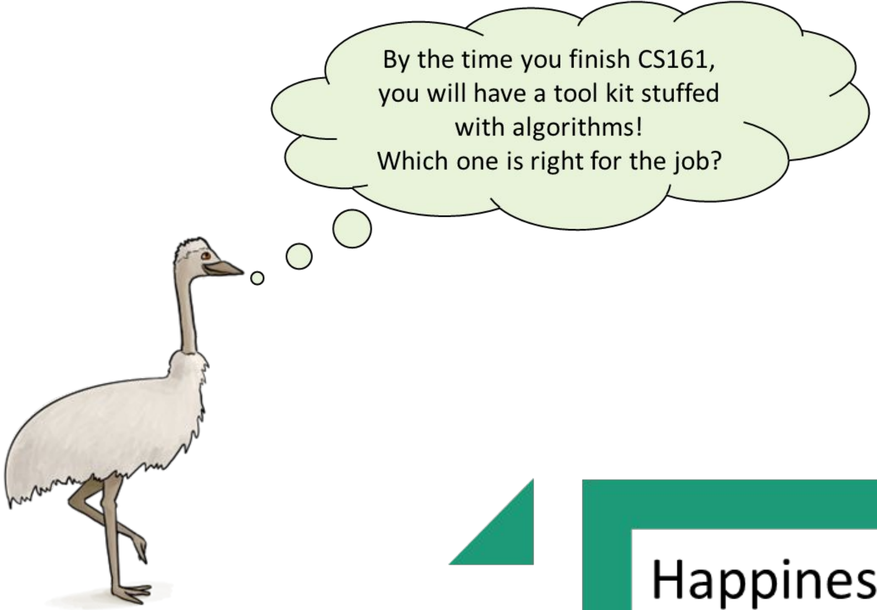
How do we make sure we aren't losing important features of the real world problem when we formalize it?



Turn a real world problem into a formal (math) problem

Use an algorithm to solve the problem

Happiness ensues



Turn a real world problem into a formal (math) problem

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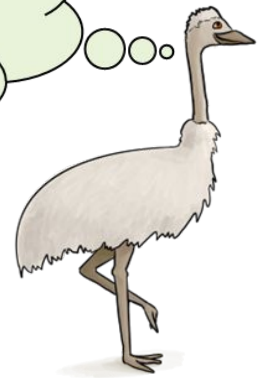
Disclaimer:
happiness not a
guaranteed outcome



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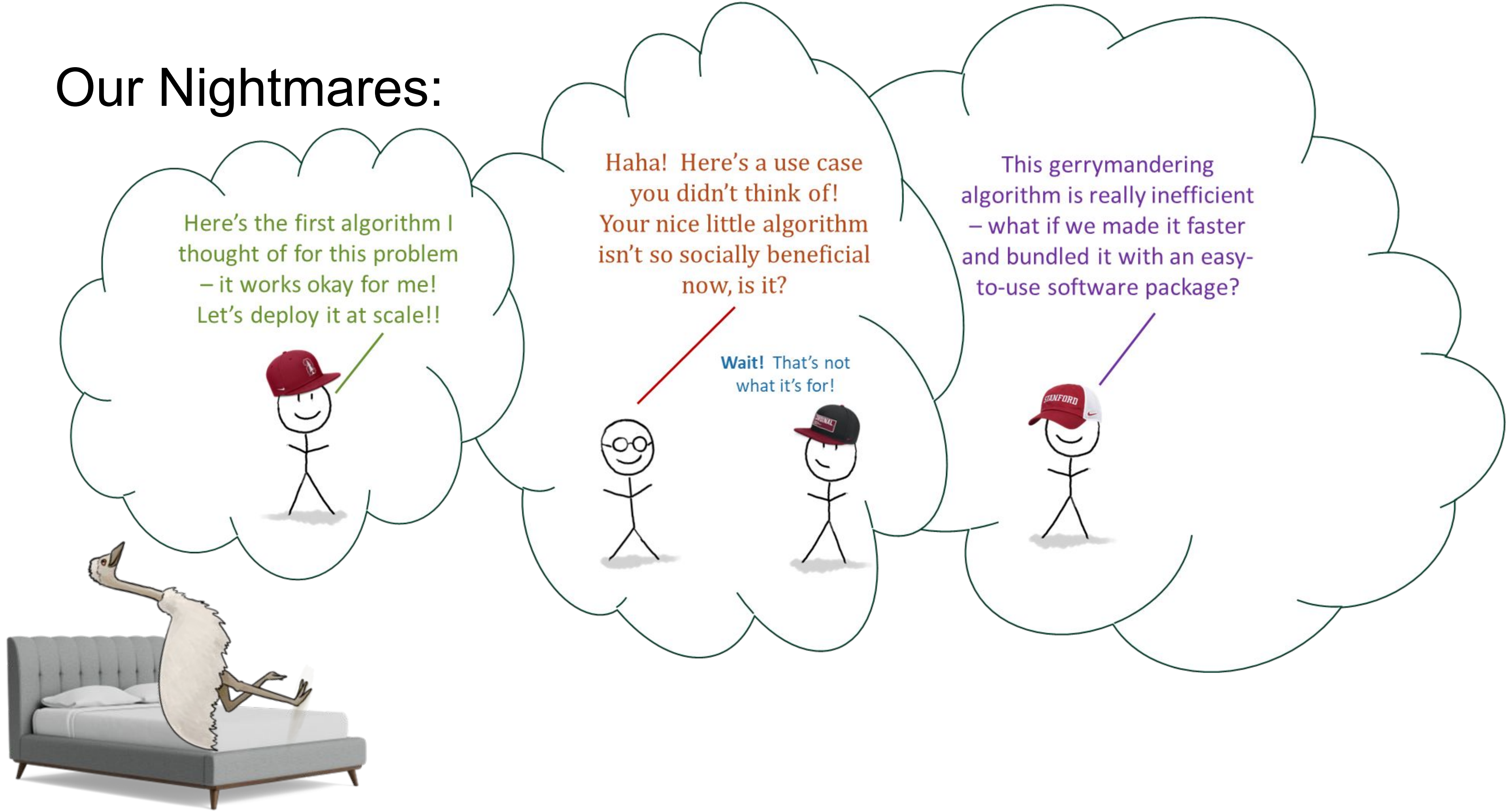
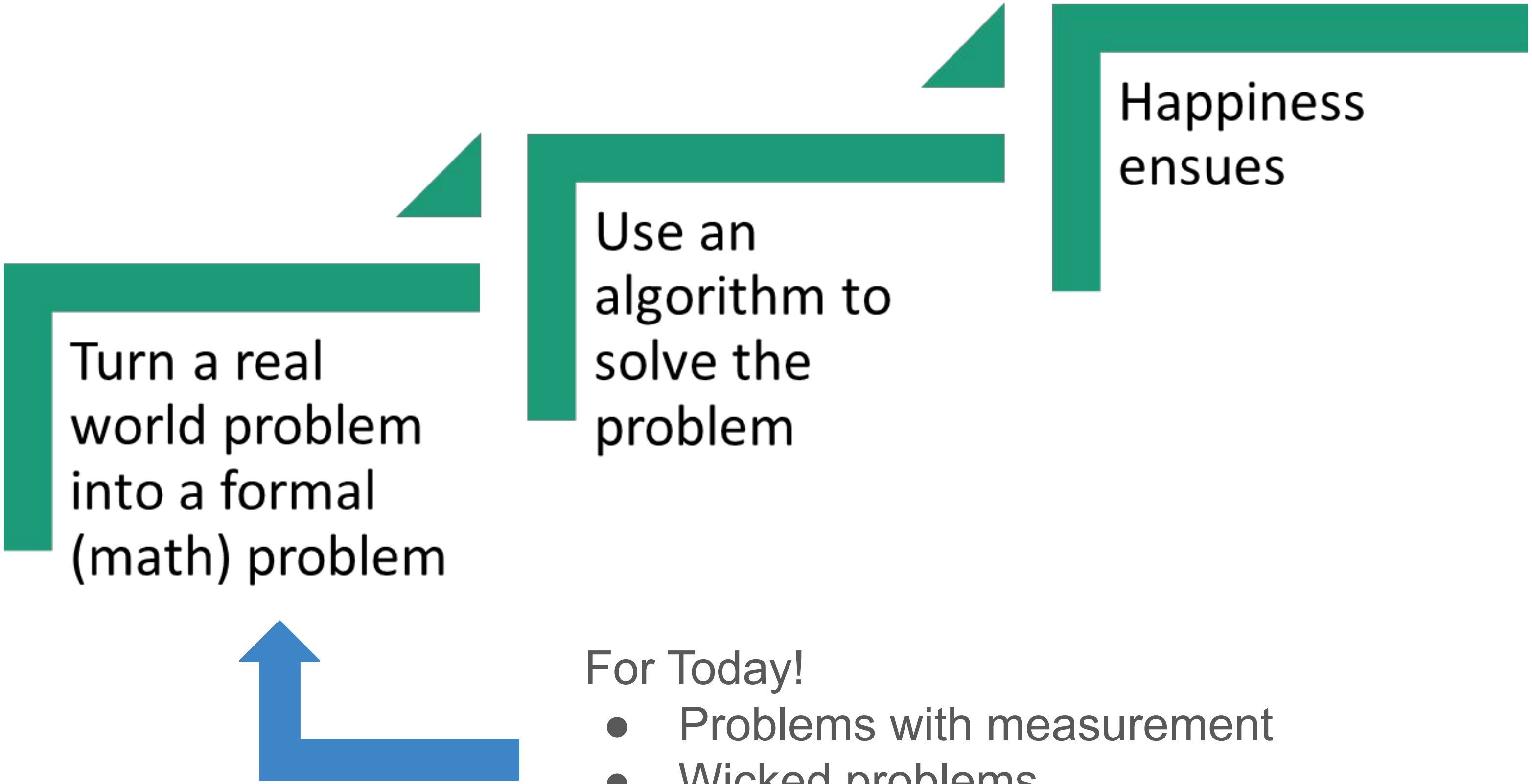
Our Nightmares:



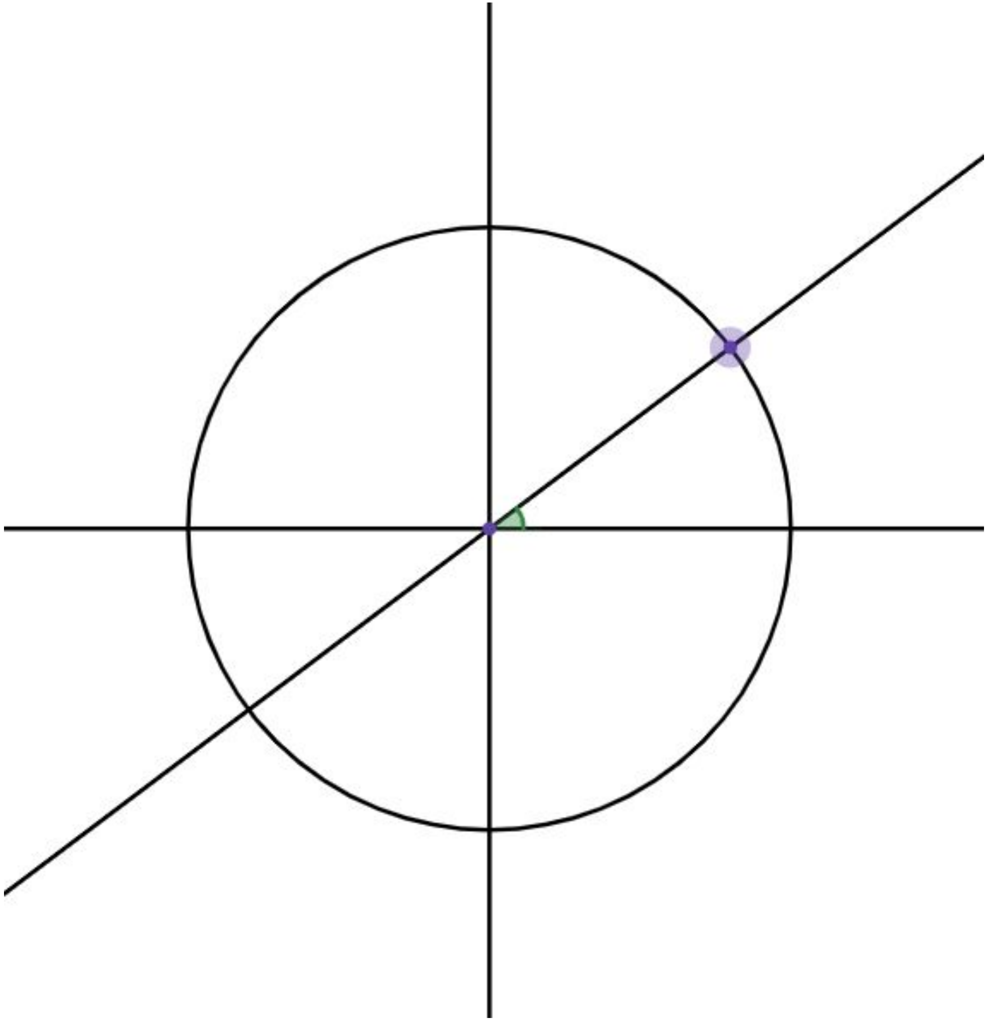


For Today!

- Problems with measurement
- Wicked problems
- Discretion
- Case studies

Problem Formalization…

Prove that if you point at the North Star with one hand, and at the Horizon with the other, the angle your arms make gives you your latitude on Earth.



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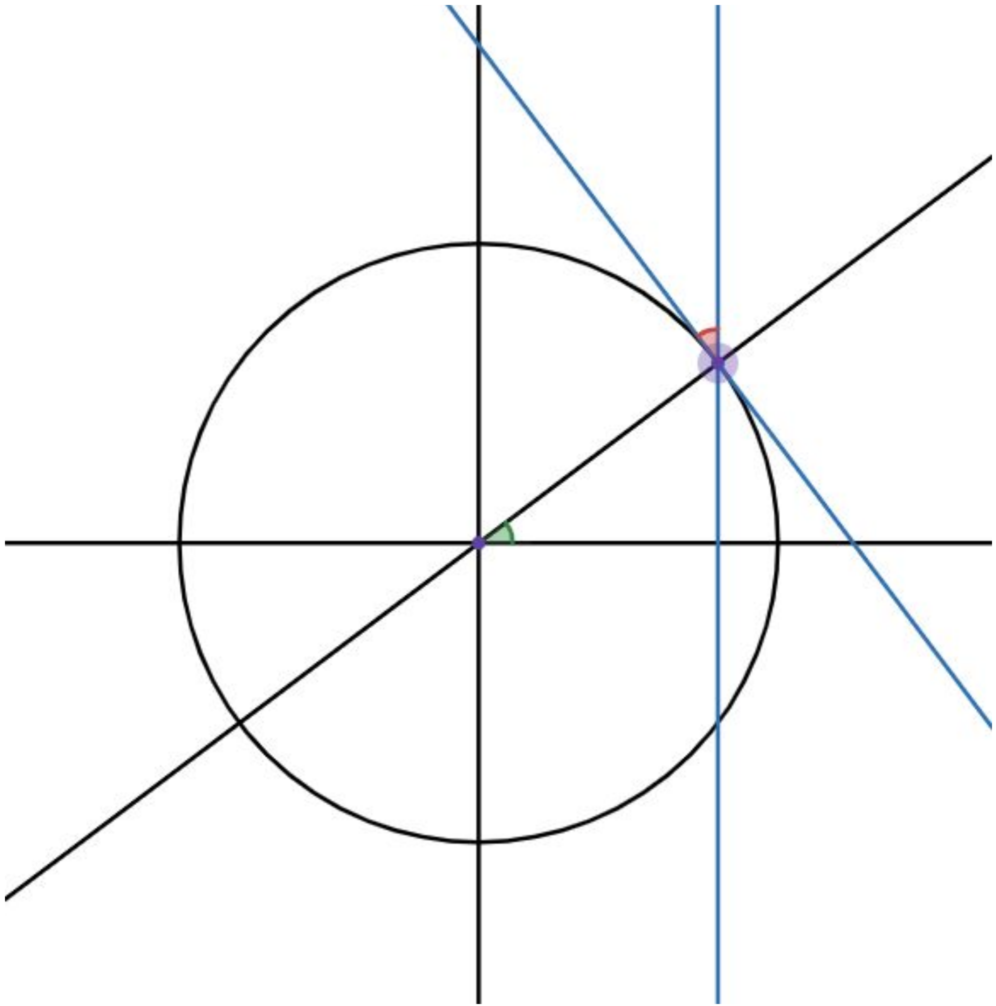

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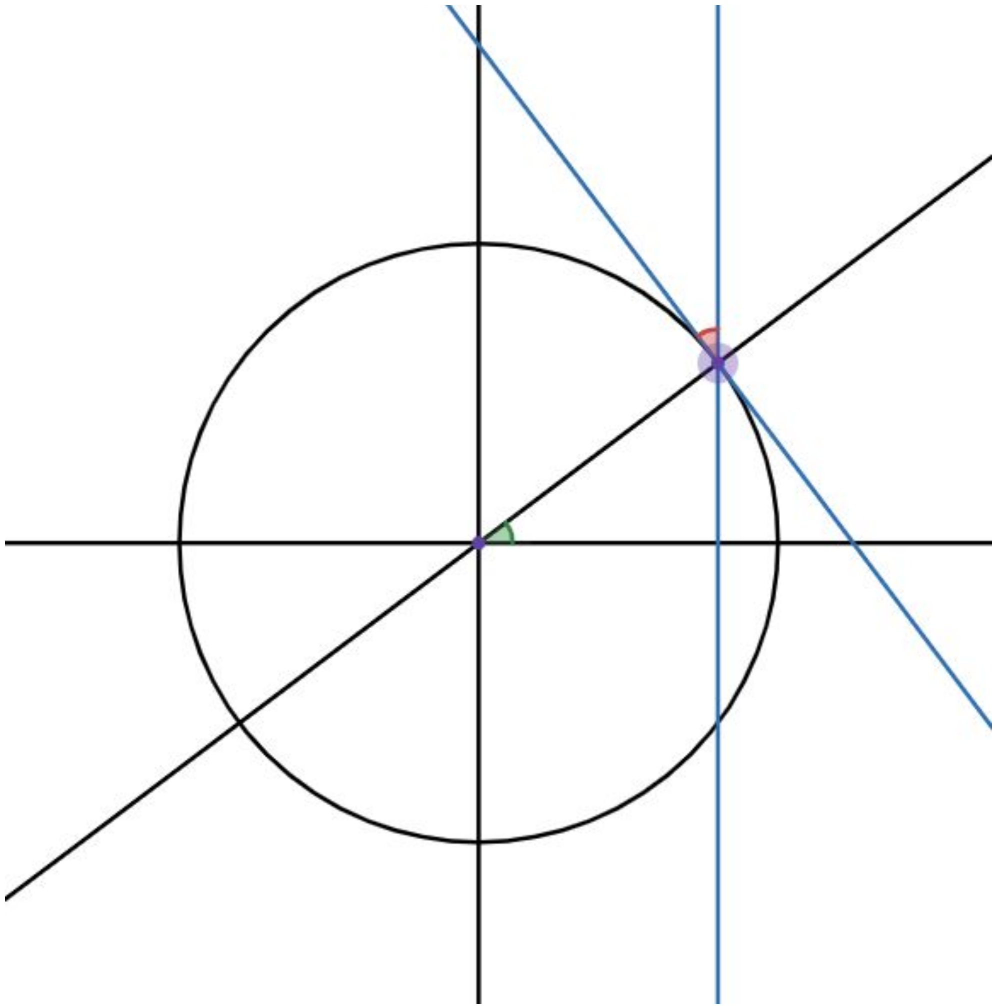

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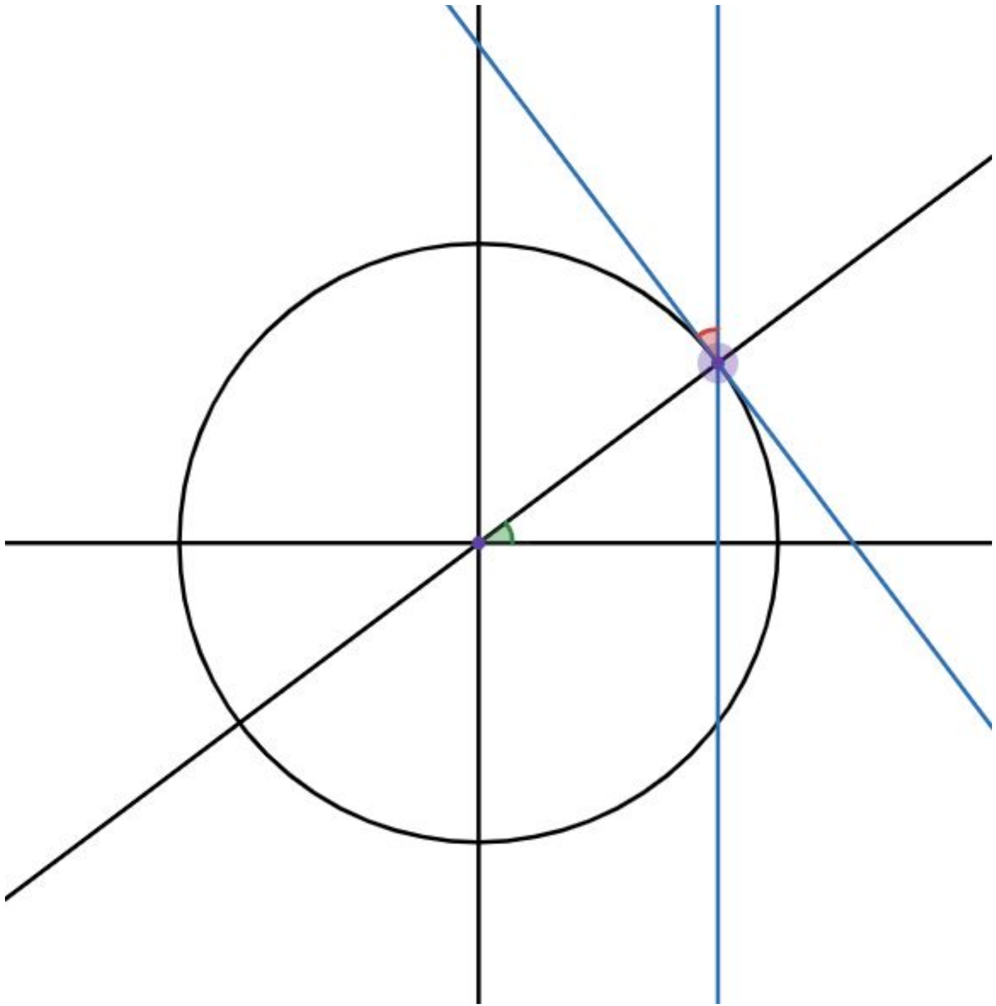

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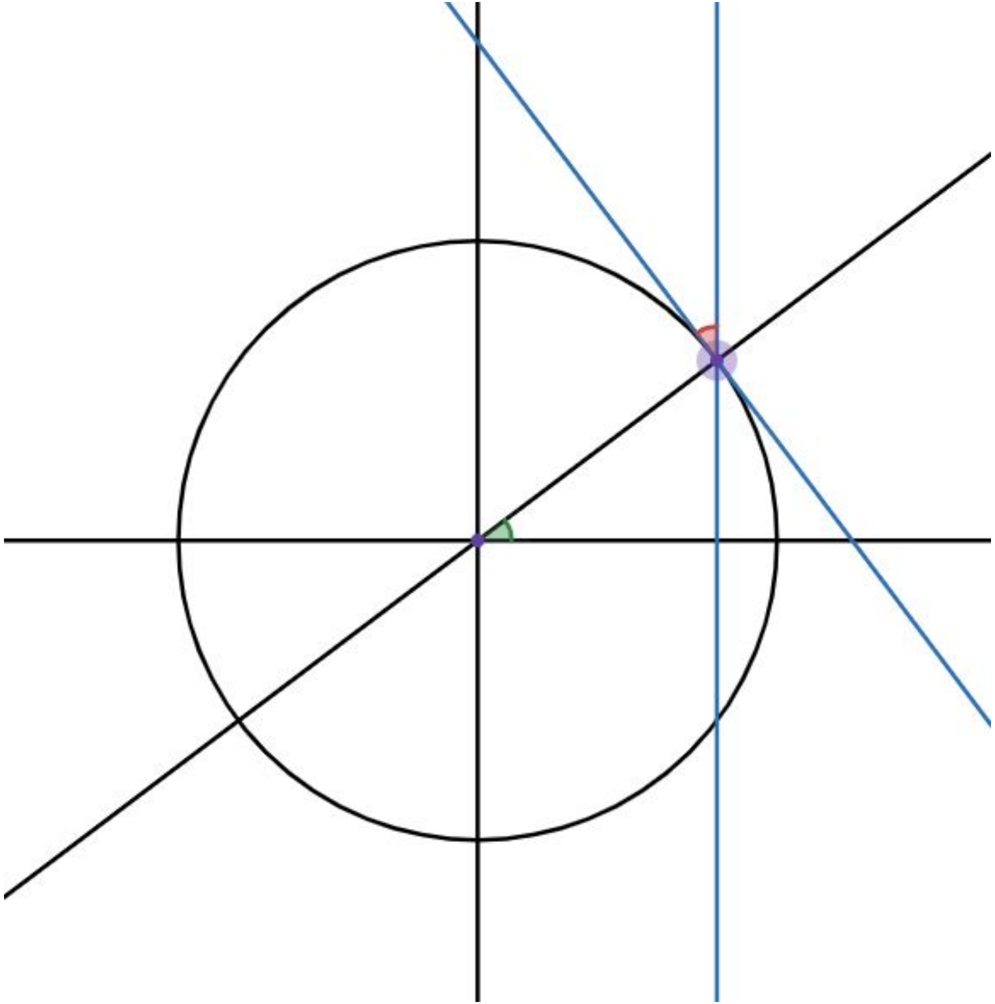

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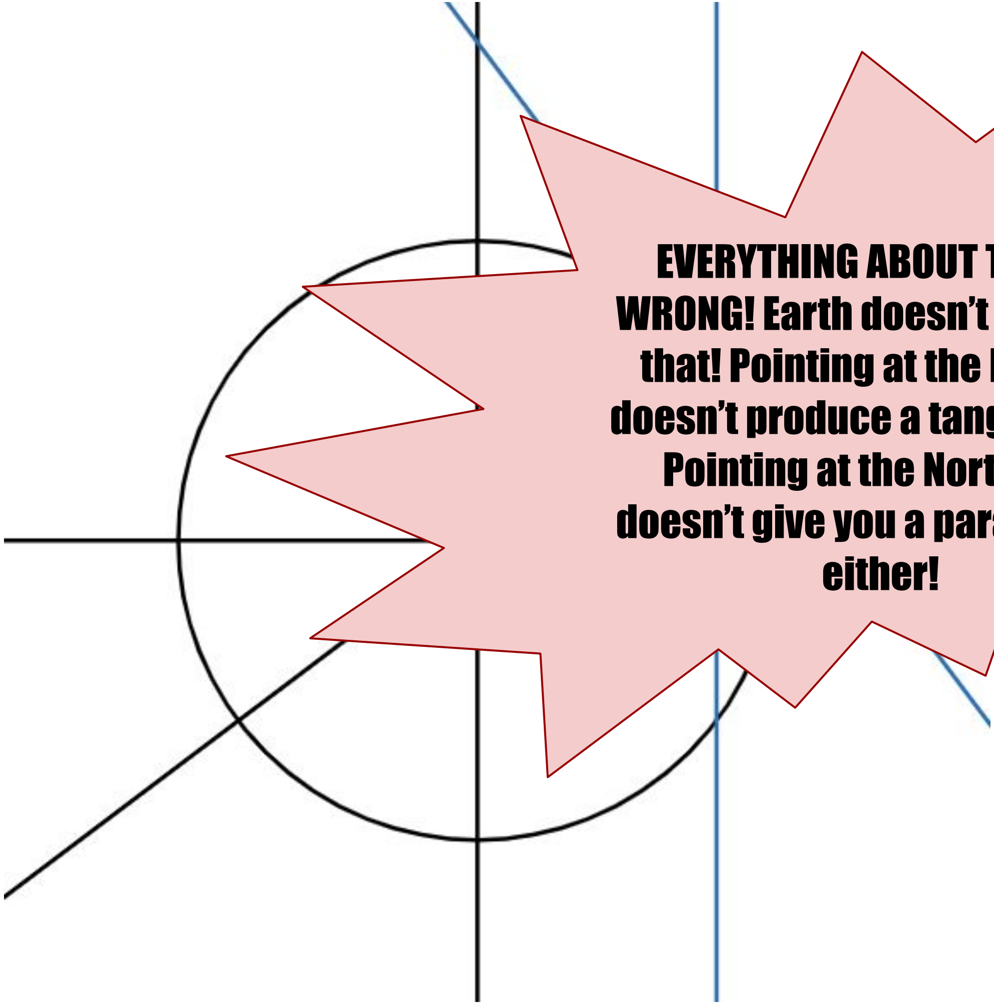


EVERYTHING ABOUT THIS IS WRONG! Earth doesn't look like that! Pointing at the Horizon doesn't produce a tangent line! Pointing at the North Star doesn't give you a parallel line either!

if you point at the ... hand, and at ... with the other, the ... make gives you ... on Earth.

- **Abstraction** is when we omit details of the real-world situation.
- **Idealization** is when we deliberately change aspects of the real-world situation.

OK, I see how abstraction and idealization might be helpful, and sometimes necessary, but it seems dangerous! What if we abstract or idealize away something important?

Measurements vs. The World

Often the value that is feasible to measure is an imperfect proxy for the thing we would actually like to measure

Examples?

GDP (Gross Domestic Product)

- The GDP is the total value of all goods and services produced within a country
- GDP is “easy” (challenging but tractable) to measure
- What should be included that is left out? Is there anything included that should be left out?

GDP as a Measure of Economic Health

As a measurement, it has three kinds of problems:

- Abstraction?
- Idealization?
- Imperfect proxy for the thing we really wanted to measure, which might have other values not captured

Route Planning Measurement Case Study

- Bogotá's Ministry of Transport wanted to measure average speeds on a road in order to do better route planning
- They set up two Bluetooth sensors spaced apart on a roadway. By picking up cell phone identifiers, Bluetooth sensors can determine the average speed of the vehicle containing that cell phone.

Route Planning Measurement Case Study

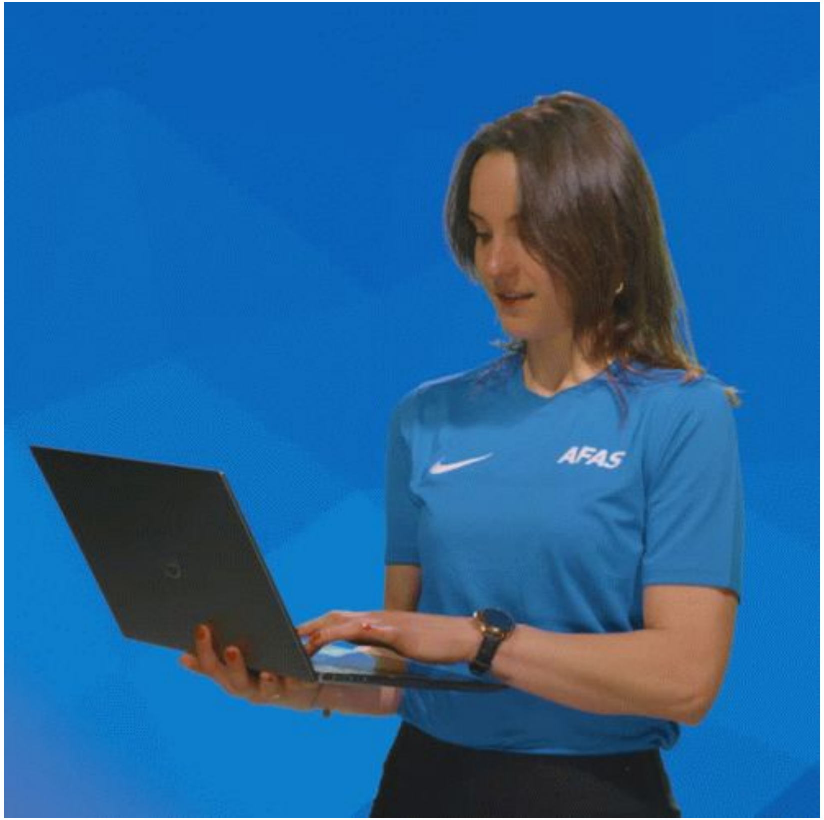
- Low-tech transport?
- Types of vehicles?
- Capacity? Multiple devices?

Abstraction and Idealization can happen in the process of formalizing a problem. A common problem spot is deciding how to measure features of a problem.

Who is included and excluded in problem formalization?

- Idealization can both include things we should exclude, and exclude things we should include
- Problematic simplification can appear even before we attempt to formulate a solution, it can appear in the data we use to detect problems in the first place

Inclusion from Data



Risks of Inclusion:

- Violation of privacy
- Data scraped without consent
- Risks to Fourth Amendment rights (self-incrimination)
- Risks to other civil liberties
- Misinterpretation of data

Exclusion from Data

Risks of Exclusion:

- Data does not describe you
- Your needs not measured/accounted for
- Voice less heard in politics
- Excluded from marketplaces/commerce
- State might take action to make you more “legible”

Case Study: Sex Categorization and Airport Security Scanners

1 A Simplification

Intentional or not, problems often begin with a simplification of the real-world. It can start with an idealization or abstraction.

2 Failed Solution Translation

Solving the simplified problem may then not translate into a practical solution for the real-world

3 Injustice

What was initially meant to be a helpful intervention leads to mistreatment or the perpetuation of injustice.

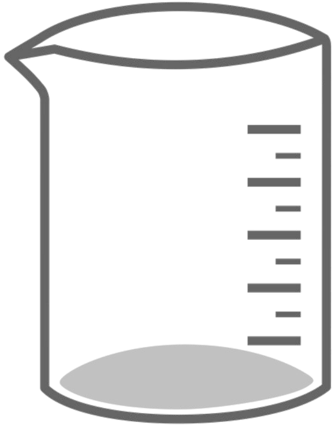
Help Me With My Next Class…

One of the classes I taught was *Mathematics and Culture*. It satisfied both the quantitative reasoning requirement and the humanities requirement. Two groups of students…

1. Math-focused students trying to fulfill the humanities requirement.
2. Humanities-focused students trying to fulfill the math requirement.

What should a grade reflect? How should I setup grading for this course? Assignments include:

1. Math problem sets
2. In-class discussion of readings
3. Final Exam with both math problems and short essay questions
4. Short papers



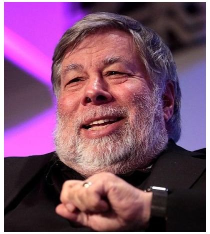
Incommensurability

Two things are *incommensurable* if they lack a *common measure* of value—that is, if there is no single measure that can be used to evaluate both.

Incommensurability makes it difficult to establish measure relationships such as “more than” or “less than, “better than” or “worse than” **by how much**.

For example, what is the relationship between milliliters (ml) and centimeters (cm)? How would you rank different amounts of water and physical lengths in the same list? You wouldn’t!

Some things are *commensurable*



- Apple stock and Microsoft stock are commensurable in terms of *dollars*: we can measure and compare them by dollar value

- Touchdowns, field goals, PATs,and safeties are commensurable in terms of *points*: we can measure and compare them by how many points they're worth

When two things are commensurable, they can easily be reduced to a single value

- f(AAPL, MSFT) = (167.43*AAPL) + (329.84*MSFT)
  - This function reduces Apple and Microsoft stock to a single dollar value

- f(TD, FG, S, EP, TPC) = (6*TD) + (3*FG) + (2*S) + (1*EP) + (2*TPC)
  - This function reduces touchdowns, field goals, safeties, extra points, and two-point conversions to a single point value

Not so with incommensurable values

What's the function for reducing income and friendships to a single number?

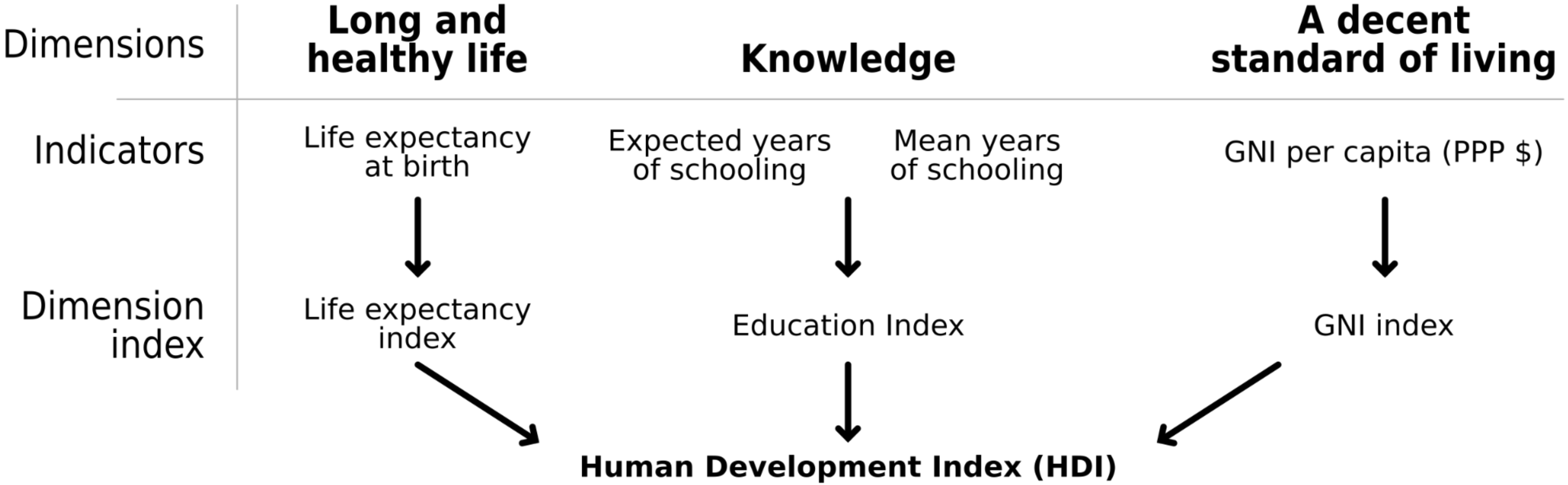
Or oil and biodiversity?

Or… time and carbon emissions?

If we are designing an algorithm where many values need to be considered, such as transportation infrastructure, it is not obvious how to compare these values against one another.

An alternative to GDP?

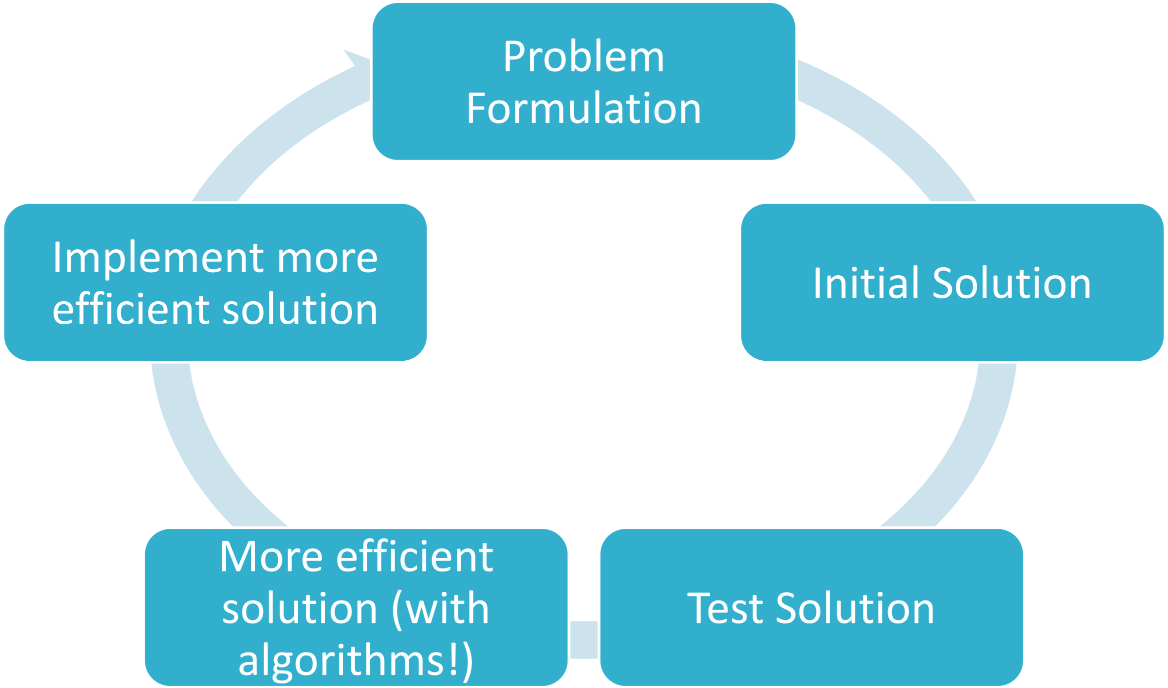
The Human Development Index:



Tame Problems

- In some ways, the problems we have addressed so far in class are tame problems
- Tame problems can still be incredibly difficult to solve!
- But at least they are tractable to formalize and formulate

Problems that are easier for computer scientists to address



“Wicked” problems

Not wicked like the Wicked Witch of the West

Instead, think: wickedly difficult

What makes a “wicked” problem wickedly hard

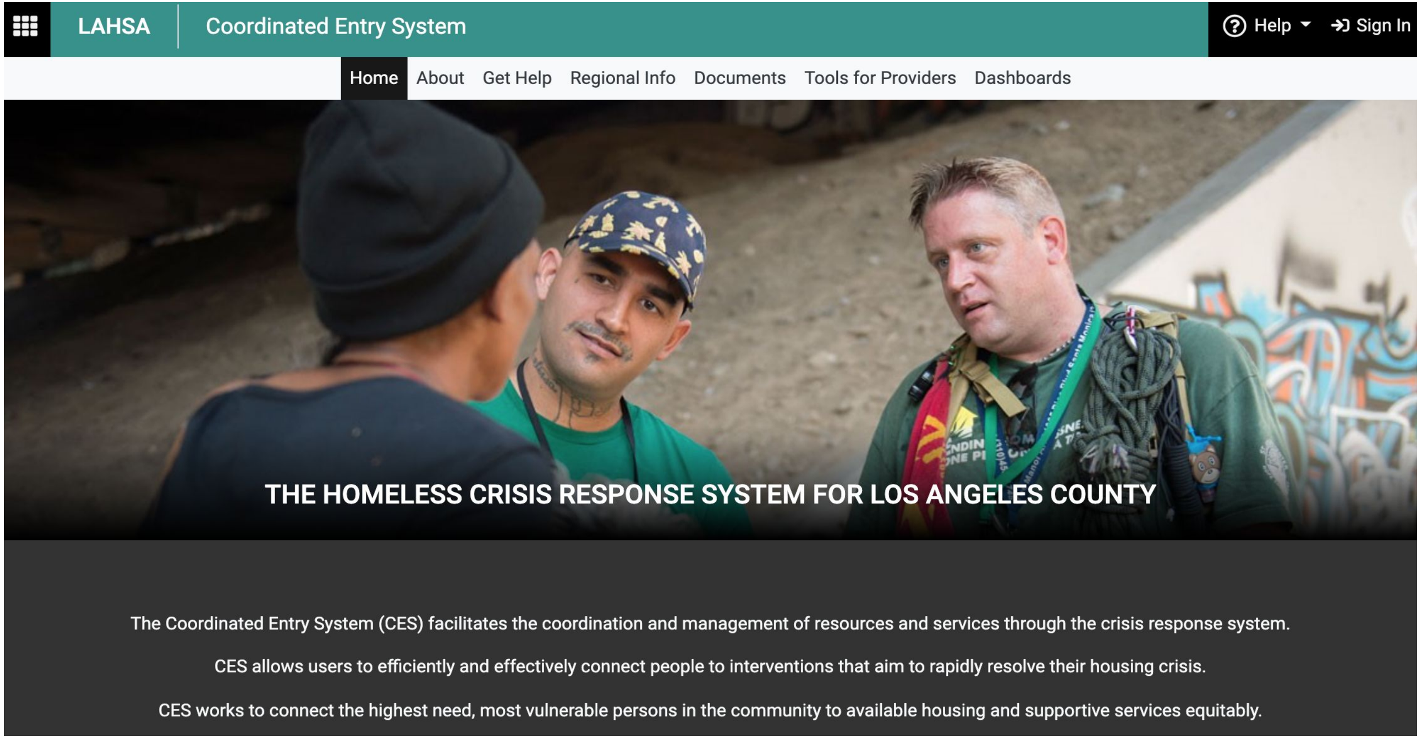
1. There is no definitive formulation of a wicked problem
2. Wicked problems have no stopping rule
3. Solutions to wicked problems are not true-or-false, but good-or-bad
4. There is no immediate and no ultimate test of a solution to a wicked problem
5. Every solution to a wicked problem is a "one-shot operation"; because there is no opportunity to learn by trial-and-error, every attempt counts significantly
6. Wicked problems do not have an enumerable (or an exhaustively describable) set of potential solutions, nor is there a well-described set of permissible operations that may be incorporated into the plan
7. Every wicked problem is essentially unique
8. Every wicked problem can be considered to be a symptom of another problem
9. The existence of a discrepancy representing a wicked problem can be explained in numerous ways. The choice of explanation determines the nature of the problem's resolution
10. The planner has no right to be wrong

Case: Homelessness

- There is no definitive formulation of a wicked problem
    - What are two different problems that could each be called “the problem of homelessness”?
- There is no stopping rule
    - When is the problem of homelessness solved?
- Solutions to wicked problems are not true-false but good-bad or better-worse
    - Some proposed solutions to homelessness are better or worse than others, but none of them stand out as “true”

Case: Homelessness

- There is no immediate and no ultimate test of a solution to a wicked problem
  - What test could one perform in order to determine that homelessness had been solved?
- Every solution is one-shot
  - Each experiment affects real people. Improving the solution later doesn't change the effect on that time in their lives.
- Every wicked problem can be considered a symptom of another problem
  - Is homelessness caused by a housing shortage? By lack of public housing? By lack of mental health treatment? We could offer many possible causes, each of which has proponents.



Case: Los Angeles County Coordinated Entry System (CES)

An electronic registry of unhoused persons who are applying or have applied to housing support programs offered by Los Angeles County.

How does it work?

Unhoused person provides (very) personal information including name, DOB, immigration status, current & past mental health, sexual activity, substance.

Algorithm uses personal data to assign a number from 1-17, least vulnerable to most vulnerable.

Risk score is used to sort by vulnerability, rank the applications, and assign housing and housing related services to the most and least vulnerable.

Question: Which problem is this solving?

Street Level Bureaucrats

Civil servants who have direct contact with members of the public and interpret and apply laws and policies. Such as

- Police officers
- Teachers
- Social Workers
- Judges
- Evaluators of welfare applications

Street Level Bureaucrats

Street level bureaucrats rely on their *discretion* in order to interpret the rules in “edge cases,” novel circumstances, and extenuating circumstances.

“street– level bureaucrats ... at least [have] to be open to the possibility that each client presents special circumstances and opportunities that may require fresh thinking and flexible action.” (Lipsky 45)

Example: Car or House?

The Carney Case… provide a procedure that determines whether or not something is car-like (subject to warrantless search) or house-like. (requires a warrant to search) Officers have much more discretion on searching a car.

Street Level Algorithms

Street level algorithms are algorithmic systems that “directly interact with and make decisions about people in a sociotechnical system. They make on–the–ground decisions about human lives and welfare, filling in the gaps between platform policy and implementation, and represent the algorithmic layer that mediates interaction between humans and complex computational systems.”

Examples:

- Resume Screening
- Content Moderation
- Bail recommendation systems

Discretion & Algorithms

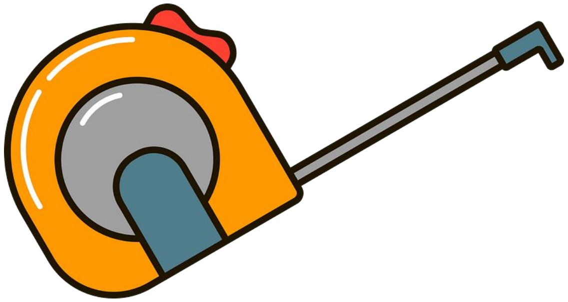
- Algorithmic decision-making systems lack discretionary judgement.
- Their decision boundaries are typically fixed
- Their procedures can be revised, but only after the decision is made incorrectly
- Q: What are possible benefits of this lack of discretion? What are possible downsides?

Case: Prerequisites

Have you or a friend ever asked a teacher or professor for permission to take a course despite not having the prerequisites? Or asked a program advisor for permission to waive a requirement?

- What would be a good reason for saying yes?
- In what case might saying yes uphold the mission of the institution even though it is against the rules?
- How should a professor exercise *discretion* when considering whether to waive prerequisites/requirements?

Measurements vs. The World



Often the value that is feasible to measure is an imperfect proxy for the thing we would actually like to measure.

Example: GDP

- The GDP is the total value of all goods and services produced within a country, easy to measure, but does it *really* measure economic health?

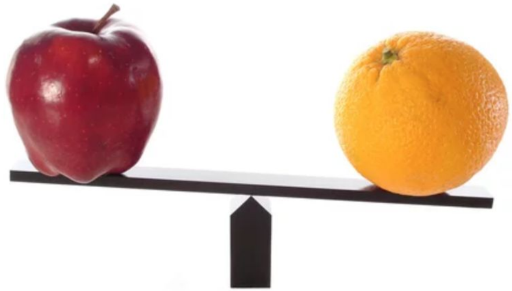
Problems with Measurements:

- ***Idealization***: Imperfect counting of the thing it measures (what production & work is it missing?)
- ***Abstraction***: counting all 10 million dollar expenditures as equally contributing to economic health (what kinds of expenditures might be more or less valuable?)
- Imperfect proxy for the thing we really wanted to measure, which might have other values not captured

Incommensurability

Two things are *incommensurable* if they lack a *common measure* of value—that is, if there is no single measure that can be used to evaluate both.

When two things are incommensurable, we *might* be able to say which is better or worse, but we can't say **by how much** (because there is no measure common to both things!)



Can you name two factors relevant to our high-speed rail problem that are incommensurable? How does their incommensurability make it hard for us to find a single optimal rail route?

Data Inclusion vs. Exclusion

Data Inclusion: When large amounts of data about you are collected

Data Exclusion: When the collected data *doesn't* include data about you

Risks of Inclusion:

- Violation of privacy
- Data scraped without consent
- "Consent" not informed
- Risks to Fourth Amendment rights (probable cause)
- Risks to other civil liberties
- Misinterpretation of data

Risks of Exclusion:

- Data doesn't describe you
- Your needs are not accounted for in political decision-making
- Exclusion from resources
- State might take action to include you in the data